

Input Form (IF)

Score: 4/5 — Minor gaps | Confidence: high

Benchmark: MILU | Context: North Indian Hindi-Medium Postgraduate Teacher — MCQ Evaluation Deployment

Input Form definition: Whether the input signal encoding (text, audio, image parameters) matches deployment conditions.

Each section below is followed by an evidence table. Three types of evidence are cited: **[QN]** — verbatim quotes extracted from the benchmark paper; **[WEB-N]** — web sources consulted during assessment; **[DN]** / **[DATASET-DN]** — sampled datapoints from the benchmark dataset, with an interpretation note.

Justification

Input form is well-aligned with the deployment context. MILU is exclusively text-based MCQ in Devanagari script for Hindi **[DATASET-D6, D7]**, which matches the deployment's Hindi/Devanagari text-only modality. Hindi is high-resource within MILU's 11-language scope **[Q5, Q76]**, questions with images, reading comprehension, and >4 options were excluded for uniformity **[Q33]**, and language verification used INDICLID and Unicode filtering **[Q34]**. Manual review steps **[Q32, Q36]** add quality assurance. The minor concerns are: (a) one truncated English item observed **[DATASET-D30]** indicates residual data quality noise; (b) translation-induced register mismatch in technical vocabulary may affect surface form for translated items **[DATASET-D14, D15, D23]**, though this is more accurately a content concern. No script, modality, or signal-distribution mismatch with the deployment.

ID	Evidence
Q5	"Models also perform better in high-resource languages as compared to low-resource ones." (p.1)
Q76	"The analysis also shows that models perform significantly better in high-resource languages than low-resource ones, highlighting the need for more robust multilingual strategies." (p.8)
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
Q34	"To remove incorrect language entries, we utilize a combination of INDICLID (Madhani et al., 2023) and Unicode-based filtering (Khan et al., 2024), ensuring that the questions are in the correct language." (p.4)
Q32	"Initially, we manually review a large sample of questions to detect and eliminate potential sources of noise." (p.4)
Q36	"As a final step, we manually verify a sample of questions from each language to ensure accuracy and correct any remaining errors." (p.4)
D6	दिए गए वाक्य का सही अपरत्यक्ष रूप चुने। प्रबंधक ने अपने सहायक से कहा कि उसे अगले महीने बोनस मल्लिगा। — Hindi grammar (indirect speech) — formal Khari Boli register
D30	The owner of the textile shop brought a ... Calculator — Truncated/incomplete question — possible data quality issue
D14	स्थिर धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
D7	'मानसून' शब्द की उत्पत्ति किस भाषा से हुई है? अरबी भाषा — Hindi-language item about etymology of 'monsoon' — GK-style
D15	रक्त में नमिनलखिति में से किसकी अधिकता 'ब्लू बेबी सिंड्रोम' रोग का कारण बनती है? नाइट्रेट — Health science MCQ — translated, with 'नाइट्रेट' transliteration retained
D23	फॉरट्रान 77 के फक्सिड फॉर्मेट में कॉलम 2 से 5 में संख्या का क्या उद्देश्य होता है? एक जंप लेबल या फॉर्मेट लेबल — Computer science MCQ — translated, technical vocabulary

Strengths

- Clean Devanagari script throughout the Hindi sample with formal Khari Boli register **[DATASET-D6, D7]**
- Image-based, reading-comprehension, and >4-option items explicitly excluded for uniformity **[Q33]**
- Language verification via INDICLID and Unicode filtering ensures script correctness **[Q34]**
- Hindi is high-resource within MILU's scope, favouring this deployment **[Q5, Q76]**

ID	Evidence
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D6	दिए गए वाक्य का सही अप्रत्यक्ष रूप चुने। प्रबंधक ने अपने सहायक से कहा कि उसे अगले महीने बोनस मलिंगा। — Hindi grammar (indirect speech) — formal Khari Boli register
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Checklist

Checklist item definitions:

ID	Assessment Question
IF-1	Compare signal distributions between source and target
IF-2	Check if regional infrastructure supports data capture specs
IF-3	Identify domain-specific form differences
IF-4	Document form mismatches harming external validity

Checklist responses:

IF-1: Signal distribution alignment is strong: both benchmark and deployment are text-only Hindi/Devanagari MCQ [Q22, Q33, DATASET-D6]. No modality, resolution, or sensor mismatch.

IF-2: Regional infrastructure can support text-only Hindi MCQ — Devanagari IME and mobile rendering are mature. Connectivity is the main constraint (Bihar 43%, UP 46% general-population internet penetration [WEB-4]; rural school internet at 18% [WEB-5]) but does not affect input-form fidelity per se.

IF-3: MCQ four-option format with single correct answer matches deployment requirement exactly [Q33, DATASET-D1]. Translation-induced terminology choices [DATASET-D14, D15, D23] are a register concern but are scored under input content rather than form.

IF-4: Minor form-level issue: one truncated English item observed [DATASET-D30], suggesting residual noise despite filtering. No systemic form mismatch.

ID	Evidence
Q22	"MILU is a large, multi-domain test set containing multiple-choice based questions (MCQs) taken from over 41 subjects with an emphasis on India-specific knowledge." (p.3)
Q33	"During the collection process, we exclude any reading-comprehension-style questions, images-based questions, and those with more than four answer options to ensure uniformity and consistency." (p.4)
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WEB-4	Bihar internet penetration ~43%, UP ~46% (IAMAI/Kantar 2024) — relevant to deployment but not benchmark form (https://www.iamai.in/sites/default/files/research/Kantar_%20IAMAI%20report_2024_.pdf)
WEB-5	Rural school internet access ~18% — affects deployment delivery, not benchmark input form (https://www.amulyacharan.com/2024/12/17/the-digital-divide-in-education-bridging-the-urban-rural-gap/)
D1	जब एक डीसी सीरीज मोटर बना लोड के चलती है: ... मोटर की गति बहुत अधिक होती है — Engineering question about DC series motor — translated from English
D14	स्थरि धारा ट्रांसफार्मर _____ प्रकार का होता है। शेल — Engineering transformer question — translated item with 'शेल' (shell) transliteration
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Information Gaps

- Frequency of truncated/incomplete items across the full dataset (only one observed in sample)

Remediation

Gap 1: Residual data quality noise observed (one truncated English item)

Recommendation: Run a lightweight pre-deployment quality filter on Hindi items (minimum stem length, option completeness checks) to surface any analogous cases before relying on MILU scores for model selection